

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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| [Website](#)

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Education

Embry-Riddle Aeronautical University

M.S. in Aviation

Specialization in Aviation Safety Systems

Daytona Beach, FL

Aug 2025–Present

GPA 4.0/4.0

Embry-Riddle Aeronautical University

B.S. in Aeronautical Science

Minor in Air Transportation

Daytona Beach, FL

May 2023

GPA 3.85/4.0

Research Interests

Aviation Safety Analytics; Air Transportation Systems; Trajectory Data Analytics; Terminal and Low-Altitude Airspace Operations

Research Experience

Graduate Researcher

Advisor: Dr. João S. D. Garcia

School of Graduate Studies, College of Aviation

Sep 2025–Present

- ML-based prediction of aircraft taxi-out times using multi-source operational data
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- Multi-layer evaluation of AAM disaster response and government-operated public use cases
- ML-based prediction of aircraft fuel burn using ADS-B and ACARS data

Graduate Researcher

Advisor: Dr. Ryan Wallace

Center for Advanced Air Mobility

Feb 2026–Present

- LAANC/UASFM assessment of shared-airspace exposure between conventional aircraft and UAS operating altitudes
- Nationwide characterization and modeling of low-altitude aircraft traffic patterns
- ADS-B-based assessment of shared airspace and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[2] Yoo, T., & Garcia, J. S. D. (Under review). Trajectory Analysis of Wind Exposure and Approach Stability in Flight Training Operations.

[1] Yoo, T., & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. *AIAA AVIATION 2026 Forum*. American Institute of Aeronautics and Astronautics.

Technical Reports

Rodrigues, E., Yoo, T., Pleffken, D. R., Cerqueira, C. S., Racharla, C. V., Crouse, S., Mendonca, F. A. C., Wadler, J., Adkins, K., Ehrenkauf, T., Keebler, J., & Garcia, J. S. D. (2026). *Synthesis framework: A practical tool linking use case features to trust and acceptance outcomes, to inform future pilot projects and public engagement strategies*.

Research Presentations

INFORMS Annual Meeting (Oral) – Accepted Institute for Operations Research and the Management Sciences <i>From Wind Exposure to Go-Arounds: The Mediating Role of Approach Stability in Commercial Aviation</i>	November 2026 San Francisco, CA
Research Match Day (Poster) Embry-Riddle Aeronautical University <i>Spatiotemporal Analysis of Aircraft Penetration Below UAS Facility Map Ceilings</i>	September 2026 Daytona Beach, FL
UAA 78th Collegiate Aviation Education Conference (Oral) University Aviation Association <i>The Effect of Wind Exposure on Go-Around Initiation and the Mediating Role of Approach Stability in Flight Operations</i>	September 2026 Denton, TX
AIAA AVIATION 2026 Forum (Oral) American Institute of Aeronautics and Astronautics <i>Data-Driven Framework for Assessing Wind Impacts on Approach Stability</i>	June 2026 San Diego, CA
Discovery Day (Poster) Embry-Riddle Aeronautical University <i>Characterizing Air Traffic Density Using Nationwide ADS-B Data</i>	April 2026 Daytona Beach, FL
COA Fall Research Colloquium (Oral) College of Aviation, Embry-Riddle Aeronautical University <i>Approach Stability Assessment Using ADS-B Data in Flight Training</i>	November 2025 Daytona Beach, FL

Funded Research Projects

2026 Safety Research Seed Grant Program Graduate Research Lead PI: João S. D. Garcia <i>An Empirical Analysis of Go-Around and Collision Risk in Support of In-Time Aviation Safety Management Systems</i>	Boeing Center for Aviation and Aerospace Safety Jul 2026–Jun 2027
Research Activities Supporting AAM National Strategies Graduate Research Assistant PI: João S. D. Garcia <i>Advanced Air Mobility in Disaster Response and Other Functions of Government as a Driver for Public Acceptance</i>	Florida Department of Transportation Feb–Jun 2026

Teaching Experience

Graduate Teaching Assistant Supervisor: Dr. Dahai Liu Teaching support for: - Statistical Analysis for Aviation/Aerospace - Research Methods in Aviation/Aerospace - Graduate Capstone Research Project - Thesis	Embry-Riddle Aeronautical University Dec 2025–Present
Flight Instructor Instructional areas include: - Private Pilot Certificate - Instrument Rating - Commercial Pilot Certificate - Certified Flight Instructor and Certified Flight Instructor-Instrument - Flight Review and Instrument Proficiency Check - Complex and High Performance Endorsement	Aviators Academy Nov 2023–Jul 2025

Honors & Awards

3 rd Place, Student Research Poster Competition, University Aviation Association	September 2026
Best Poster Award, Embry-Riddle Aeronautical University	April 2026
1 st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University	November 2025
Magna Cum Laude, Embry-Riddle Aeronautical University	May 2023
Dean's List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University	

Certifications & Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane
Flight Instructor – ASEL, AMEL, Instrument Airplane
Ground Instructor – Advanced, Instrument
Remote Pilot – sUAS

Technical Skills

Python, SPSS, LaTeX

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS)
American Institute of Aeronautics and Astronautics (AIAA)
University Aviation Association (UAA)
Royal Aeronautical Society (RAeS)

Leadership & Service

Mentor	American Institute of Aeronautics and Astronautics
AIAA Cape Canaveral Section Mentorship Program	2026–Present
Secretary	Embry-Riddle Aeronautical University
Korean Student Association	2019–2020
Administrative Specialist; Squad Commander	Korea Defense Communication Command
Republic of Korea Military Service	2017–2019
Vice Captain	University of North Dakota
Table Tennis Club	2016–2017